

C++2: Lesson 6 – Lab Project

1. Add **summary** at the top and **comments** where necessary.
- Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

2

100 points

Use an **unordered_map** to manage the certificates:

1. Create a **Student** class to store the **name** and **certificate** of a student.
2. Write a test code to **input** the information below and **print** the names of the students who have the specified certificate.

Name	Certificate
Henry	Java
Ben	Python
Ava	C++
Jane	Java
Emma	Python
Rick	C++

Sample
output

```
Certificate? C++
Student(s) with a C++ Certificate
-----
Ava
Rick

Certificate? Ruby
No students found with a Ruby Certificate
```

File(s) to
submit

C206_2.cpp
C206_2.png

3

100 points

Write a program to:

1. **Prompt** the user to enter the number of random integers to generate (positive integer).
 2. **Generate** the specified number of **random integers** between 0 and 9.
 3. **Calculate** the mode (the most frequently occurring number).
-If multiple integers have the same highest frequency, display any one of them as the mode.
 4. **Display** the mode and its frequency.
- *Choose the most appropriate data structure from the set types.*

Sample
output

```
Integer? 20
Generated integers: 6 3 7 1 8 3 5 9 0 7 2 7 1 8 0 5 9 6 4 1
Mode: 7 (appears 1 times)
```

File(s) to
submit

C206_3.cpp
C206_3.png